

Challenge (Habanero)

Q1. What is the discriminant of $x^2 + 5x + 6 = 0$?

- (A) 1
(B) 13
(C) -11
(D) 25
(E) -1

Q2. A gamer's score S is given by $S = t^2 - 4t + 4$, where t is time. How many real solutions are there for $S = 0$?

- (A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q3. Solve $x^2 + 2x - 6 = 0$ using the quadratic formula. What is the value of x ?

- (A) $-1 \pm \sqrt{7}$
(B) $1 \pm \sqrt{7}$
(C) -3, 2
(D) 2, -3
(E) -2, 3

Q4. The cost C of running a gaming server is given by $C = x^2 - 7x + 12$. What is the minimum cost and when does it occur?

- (A) $C = 12, x = 0$
(B) $C = 4, x = 3$
(C) $C = 1, x = 4$
(D) $C = 0, x = 7$
(E) $C = -1, x = 6$

Q5. A coding logic puzzle involves solving $2x^2 - 5x - 3 = 0$. What is the discriminant?

- (A) 1
(B) 49
(C) -29
(D) 17
(E) -1

Q6. The score S of a player in a game is modeled by $S = -t^2 + 4t + 3$, where t is time. For what values of t is $S = 0$?

- (A) $t = -1, 3$
(B) $t = -3, 1$
(C) $t = 1 \pm \sqrt{7}$
(D) $t = -1 \pm \sqrt{7}$
(E) $t = 3, -1$

Q7. Determine the nature of the solutions for $x^2 - 2x - 15 = 0$.

- (A) Two real, distinct solutions
(B) Two real, equal solutions
(C) No real solutions
(D) Infinitely many solutions
(E) One real solution

Q8. The data usage D of a gamer is given by $D = t^2 + 2t + 1$, where t is time. How many times will $D = 0$ occur?

- (A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Open Ended Questions (FRQ)**Q9.** Solve the equation $x^2 - 3x - 2 = 0$ using the quadratic formula and interpret the discriminant. Show all steps and explain your answer.**Q10.** The cost C of running a gaming server is given by $C = x^2 - 5x + 6$. Determine the minimum cost and when it occurs. Show all steps and explain your answer.**Chef's Tips**

- Check calculations carefully
- Consider all possible solutions
- Use the quadratic formula when necessary